

 UTM UNIVERSITI TEKNOLOGI MALAYSIA RESEARCH UNIVERSITY	FACULTY OF COMPUTING UNIVERSITI TEKNOLOGI MALAYSIA
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PSM 1 (SEC_3032)

PROJECT PROPOSAL & SUPERVISION CONSENT FORM

Session/Semester: ..2025/2026.. 2 ..

Instruction: Please complete and submit to your program PSM1 committee member.

Student: Please complete student information in detail. The proposal must be reviewed by the supervisor before submission.

Supervisor: Please complete your consent, signature, stamp and date. Please review student's proposal before submission.

Note: Students are ONLY ALLOWED to do Research type projects if they have a CGPA of 3.3 and above.

SECTION A: STUDENT INFORMATION

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Year/Course:	3/SECBH		
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CGPA: 3.87	Date: 02-04-2026	Signature:	

SECTION B: SUPERVISOR CONSENT

Supervisor Name:	Ts. Dr. Chan Weng Howe		
Email:	cwenghowe@utm.my	Mobile No.	016-4247363
Consent:	I AGREE to be an Undergraduate Supervisor for this student with the proposed title that I have reviewed.		
Signature & Official Stamp:	 TS. DR. CHAN WENG HOWE Senior Lecturer Faculty of Computing Universiti Teknologi Malaysia cwenghowe@utm.my	Date:	03-Apr-2026

SECTION C: PROJECT DETAILS

Project Title:	Deep Learning-Based Methods for Latent Representation Quality in Multi-Omics for Cancer Subtype Classification
Proposal No.*	1
Project Idea:	☐ My own idea ☒ Proposed by the supervisor
Project Type:	☒ Research ☐ System Development

**(Please follow your preference. Proposal No. 1 – the highest priority, followed by Proposal No. 2 Each student may propose a maximum of 2 topics).*

Problem Background and Proposed Solution:

Cancer is a highly heterogeneous disease on molecular level and therefore making accurate subtype classification are crucial for precision medicine. Multi-omics data, which combines gene expression (mRNA), microRNA (miRNA), DNA methylation, and copy number variation (CNV) offers a rich molecular view of tumour behaviour but is challenging to analyse as it is highly dimensional and represents a heterogeneous biological layer. Although numerous latent representation learning methods have been proposed, ranging from statistical techniques such as MOFA+, IntNMF, to deep learning techniques like VAE and CustOmics, those existing evaluations are inconsistent and rarely assess the quality of the learned latent representations beyond classification accuracy alone.

Thus, this study aims to evaluate latent representation quality in multi-omics cancer subtype classification with Variational Autoencoder (VAE) as the primary method of focus. Multi-omics data from The Cancer Genome Atlas (TCGA) will serve as the standardised dataset. This study will be assessed across classification performance (Accuracy, F1 score, AUC-ROC), latent space structure (Silhouette score, NMI), and biological relevance (survival correlation), providing evidence-based guidance for researchers in selecting appropriate methods for multi-omics cancer analysis.mRNA, miRNA, DNA methylation, and copy number variation(CNV).

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Objectives:

- To review existing latent representation methods for multi-omics cancer subtype classification, covering both statistical-based and deep learning-based approaches.
- To implement the Variational Autoencoder (VAE) for learning latent representations from multi-omics data.
- To evaluate the strengths and limitations of selected latent representation methods based on classification performance, latent space quality, and biological relevance.

Scopes:

This project focuses on the evaluation of existing latent representation techniques for data like multi-omics cancer data, specifically mRNA, miRNA, DNA methylation, and CNV data, using publicly available data from the TCGA database. This study is using Variational Autoencoder as the main method, along with some other methods for comparison to selected baseline methods where appropriate. While for the evaluation, it is restricted to three aspects which are classification performance, latent space structural quality, and biological relevance.

Data source : TCGA (The Cancer Genome Atlas)

Project Requirements:

Software	Programming Languages: Python Development Environment: Visual Studio Code Version Control: Git & GitHub
Hardware	Processor (CPU): 11th Gen Intel® Core™ i5-11300H @ 3.10 GHz Memory (RAM): 16 GB Storage: 477 GB Solid State Drive (SSD) Graphics Processing Unit (GPU): 4 GB Dedicated Graphics Card System Architecture: 64-bit Operating System, x64-based processor
Technology/Technique / Method/Algorithm	Latent representation learning methods: Variational Autoencoder (VAE) Data preprocessing and integration: Handling multi-omics data (mRNA, miRNA, DNA methylation, CNV) Classification techniques: Support Vector Machine (SVM) Evaluation methods: Classification performance — Accuracy, F1-score, AUC-ROC Latent space quality — Silhouette Score, Normalised Mutual Information (NMI) Biological relevance — Survival analysis using lifelines
Network Elements	Internet access required for downloading datasets from TCGA and accessing public repositories
Security Elements	• All datasets used are publicly available and de-identified (TCGA). • No patient-identifiable information is handled.
Project Area:	Bioinformatics - Cancer Genomics